In the same vein, we have

$$\begin{aligned} 2(W_U-(1-\delta)V) &= \mathrm{Tr}(\tau^{-1}\sigma_\beta)-(1-\delta)\ \log\det(2\pi\overline{\sigma})-\log\det(2\pi\tau) \\ &\quad -(y-\overline{m})'\left((1-\delta)\ \overline{\sigma}^{-1}-\tau^{-1}\right)(y-\overline{m}) \\ &\quad +2(\overline{m}-m_0)'\tau^{-1}(y-\overline{m})+(\overline{m}-m_0)'\tau^{-1}(\overline{m}-m_0), \end{aligned}$$

which shows that

$$\nu_{(1-\delta)V}(e^{W_U})<\infty \Longleftrightarrow \tau^{-1}<(1-\delta)\ \overline{\sigma}^{-1}.$$

■

**Proof of Proposition 5.4.** Let us first recall that

$$\lambda_U(dx)=\sum_{i\in E_0}\xi_0(i)\ \lambda_{U_i}(dx) \quad \text{and} \quad \nu_V(dy)=\sum_{j\in E_1}\xi_1(j)\ \nu_{V_j}(dx),$$

which implies that

$$W_U=\sum_{i\in E_0}\ \xi_0(i)\ W_{U_i} \quad \text{and} \quad W^V=\sum_{j\in E_1}\ \xi_1(j)\ W^{V_j} \tag{E.3}$$

and ensures that the functions $W_U$ and $W^V$ are locally bounded.

Moreover, we note that

$$\sum_{i\in E_0}\xi_0(i)\ e^{-U_i}\leqslant e^{-\inf_{i\in E_0}U_i},$$

which implies that

$$-U=\log\sum_{i\in E_0}\xi_0(i)\ e^{-U_i}\leqslant-\inf_{i\in E_0}U_i$$

and, hence,

$$U\geqslant\inf_{i\in E_0}U_i.$$

As a consequence, we readily obtain the inequalities

$$\delta U-W^V\geqslant\inf_{i\in E_0}\sum_{j\in E_1}\ \xi_1(j)\ (\delta\ U_i-\ W^{V_j})\geqslant\inf_{(i,j)\in(E_0\times E_1)}(\delta\ U_i-\ W^{V_j})$$

which, in turn, enable us to show that

$$\begin{aligned} \left\{x\ :\ \delta U(x)-W^V(x)\leqslant r\right\} &\subset \left\{x\ :\ \inf_{(i,j)\in(E_0\times E_1)}(\delta\ U_i(x)-\ W^{V_j}(x))\leqslant r\right\} \\ &= \bigcup_{(i,j)\in(E_0\times E_1)}\left\{x\ :\ (\delta\ U_i(x)-\ W^{V_j}(x))\leqslant r\right\} \end{aligned}$$

and this ensures that $\mathcal{U}_\delta=\delta U-W^V$ have compact sub-level sets. In the same vein, one can also prove that $\mathcal{V}_\delta=\delta V-W_U$ have compact sub-level sets.

On the other hand, using the fact that $(a+b)^{1-\delta}\leqslant a^{1-\delta}+b^{1-\delta}$ for any $a,b\geqslant 0$, we have

$$\begin{aligned} \lambda_{(1-\delta)U}(dx) &= \left(\sum_{i\in E_0}\xi_0(i)\ e^{-U_i(x)}\right)^{1-\delta}\lambda(dx) \\ &\leqslant \sum_{i\in E_0}\xi_0(i)^{1-\delta}\ e^{-(1-\delta)U_i(x)}\ \lambda(dx)=\sum_{i\in E_0}\xi_0(i)^{1-\delta}\ \lambda_{(1-\delta)U_i}(dx) \end{aligned} \tag{E.4}$$

and, by way of Hölder's inequality, we also have

$$\lambda_{(1-\delta)U_i}\left(\prod_{j\in E_1} e^{\xi_1(j)\ W^{V_j}}\right) \leqslant \prod_{j\in E_1}\left(\lambda_{(1-\delta)U_i}\left(e^{W^{V_j}}\right)\right)^{\xi_1(j)}. \tag{E.5}$$

Combining (E.3), (E.4) and (E.5) we obtain the estimates

$$\begin{aligned}\lambda_{(1-\delta)U}\left(e^{W^V}\right) &\leqslant \sum_{i\in E_0}\xi_0(i)^{1-\delta}\ \lambda_{(1-\delta)U_i}\left(\prod_{j\in E_1} e^{\xi_1(j)\ W^{V_j}}\right)\\ &\leqslant \sum_{i\in E_0}\xi_0(i)^{1-\delta}\prod_{j\in E_1}\left(\lambda_{(1-\delta)U_i}\left(e^{W^{V_j}}\right)\right)^{\xi_1(j)}\end{aligned}$$

from which we conclude that $\lambda_{(1-\delta)U}\left(e^{W^V}\right) < \infty$ (because $\lambda_{(1-\delta)U_i}\left(e^{W^{V_j}}\right) < \infty$ for almost every $(i,j)$). Using the same argument, we can also prove that $\nu_{(1-\delta)V}\left(e^{W_U}\right) < \infty$. ■

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